Communication-Aware Collaborative Robotics

The Scenario, and the Literature by Category

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1. The Scenario

A robot team on shared Wi-Fi in a changing building, keeping each robot's picture of the places its next move depends on correct, in time.

Some robots move through the building on jobs; some fixed cameras watch parts of it; all share one access point with the building's other traffic. People walk, doors open and close, things get moved. Each robot is about to act somewhere: turn a corner, enter a room, cross a junction. To act well it needs a correct picture of the places its next move depends on. Some it will see for itself in time; some are hidden and it can only learn about them from another robot or a camera that sees them now. Every observation is about a place, taken at a time, and stays true for a while: a wall for ever, a parked cart for minutes, a walking person for a second or two.

The problem the team faces at every moment is the following.

> *Of everything anyone has seen, which observations should go to which robot, in what form, sent by whom, and when, so that every robot has a correct picture of the places its next move depends on by the time it makes that move, given that the link cannot carry everything and the robots keep moving?*

Four properties make the problem hard, and they are the four axes along which the literature is organised below.

1. Deadlines come from motion. When a robot needs a place is set by how fast it moves toward it.
2. Validity comes from the world. How long an observation stays true depends on what is in it and how busy the place is.
3. Need comes from the receiver's own future. A place the robot will see itself before it matters needs no message.
4. The channel is shared and short. One access point; messages collide; bulk traffic delays small urgent messages.

The quantity. Everything is scored by the age of incorrect information at the decision (AoII at the decision): how long a robot's belief about a place has been wrong at the moment its move depends on that place.

The approach. Decide what to send by when it is needed and how long it stays true; take turns on the channel; let the robot move to keep its picture right; score at the moment it acts.

Experiments, in order of size. One blind corner (one mast, one robot, a loading robot; SCENARIO_0.md); the crossing (three masts, two robots, a throttled link; SCENARIO_CROSSING.md); the changed building (a moved cabinet, a closed door); the full rig under pose and control deadlines.

Rig. Two Scout Mini robots (Jetson Orin NX, Ouster OS1, ZED 2i, two TI radars), three Raspberry Pi masts with cameras, one Asus RT-BE82U access point, one server, chrony (measured offset at most 0.99 ms), one monitor-mode sniffer, and the reference map of site 1 preloaded on every node. No digital twin, no ray tracer, and no labels beyond who was where per trial.

2. The Literature by Category

Each category owns one of the four axes and assumes the others away. Tags: [F] read in full in this project (PDFs in the refs folder); [V] venue checked against a listing; [S] abstract or summary only, to be checked before citation. Entries without a tag are classical results whose records are well known; the journal volume and pages of the 2004 and 2008 information-theory entries were not checked here.

2.1 Collaborative perception: what to send

Owns which cells, features or objects are sent, and how few bytes. Assumes away the channel (a byte budget or a constant delay), the receiver's future, deadlines, and other senders.

Table 1. Collaborative perception.

Paper	Venue	Note
Where2comm (Hu, Fang, Lei, Zhong, Chen)	NeurIPS 2022	spatial confidence and request map; the canonical baseline
V2X-ViT (Xu et al.)	ECCV 2022	delay-aware fusion; constant 100 ms delay
CoBEVT (Xu et al.)	CoRL 2022 [S]	sparse BEV fusion
CodeFilling (Hu et al.)	CVPR 2024 [S]	codebook messages

Paper	Venue	Note
CoSDH (Xu et al.)	CVPR 2025 [S]	supply–demand masks; intermediate–late hybrid
ERMVP (Zhang et al.)	CVPR 2024 [S]	efficient and robust messages
Chiu, Smith, selective communication	2023 [S]	object centres first, detail on request
InfoCom	AAAI 2026 [S]	kilobyte messages by information bottleneck
RDComm (Liu, Hu, Hu, Zhang, Chen)	ICLR 2026 [V]	pragmatic rate–distortion; ICLR title differs from arXiv
WhisperNet (Chen, Zhang, Zhao)	CVPR 2026 [S]	sender importance and receiver coordination at 1 % bandwidth
Dense Coverage, Sparse Refinement (Yazgan, Mueller, Zoellner)	arXiv 2609.29456, 2026 [F]	coarse base plus 198-byte patches ranked by learned benefit; strict byte caps; cross-agent redundancy stated as open
JigsawComm (Wang, Li, Li, Wen)	arXiv 2511.17843, 2026 [F]	learned per-cell utility, top-1 sender per cell, O(1) payload in N; decision identical at every agent; a second sender per cell lowers accuracy
SparseCoop (Wang et al.)	AAAI 2026 [V]	instance queries with explicit position and velocity
TraF-Align (Song, Yang, Wen, Li)	CVPR 2025 [V]	trajectory-aware feature alignment under latency
CoBEVFlow (Wei et al.)	NeurIPS 2023	BEV flow for asynchrony
FFNet (Yu et al.)	NeurIPS 2023 [S]	sender-side feature flow
SyncNet (Lei et al.)	ECCV 2022	latency-aware fusion
Reason-to-Transmit (Bansal, Gangwani)	arXiv 2603.20308, 2026 [F]	transformer selection with a neighbour-state token; selection policies within about 0.5 AP of each other; fusion dominates

An indoor ground-robot benchmark with a density-guided residual fusion operator was seen in confidence while under review; it is not citable and is omitted from the table.

2.2 Task-oriented and semantic communication: how to encode

Owns bits that preserve a task loss over a noisy physical channel. Assumes away the medium-access layer (no contention, no queues), the receiver's state, and time.

Table 2. Task-oriented and semantic communication.

Paper	Venue	Note
Gündüz et al., Beyond Transmitting Bits	JSAC 2023	the survey
Shao, Mao, Zhang, task-oriented communication for edge inference	JSAC 2022	information bottleneck
Shao, Mao, Zhang, multi-device cooperative edge inference	TWC 2023	distributed information bottleneck
Lan, Zeng, Popovski, Gündüz, Huang, ProgressFTX	TWC 2022 [S]	progressive feature transmission with threshold stopping
SComCP (Gan et al.)	arXiv 2507.00895, 2025 [V]	semantic communication for collaborative perception over a PHY channel
CoDS (Gan, Liang, Zhang, Guo, Jin)	arXiv 2512.22513, 2025 [V]	digital semantic communication for collaborative perception
R-ACP (Fang et al.)	JSAC 2025 [V]	capacity-adaptive compression with priority scheduling on real vehicles
Token Communications (Qiao et al.)	IEEE Wireless Commun. 2025 [S]	tokens as the transmission unit

Foundations used by this category and by the present work: Equitz and Cover (T-IT 1991) and Steinberg and Merhav (T-IT 2004) on successive refinement, with and without side information; Tian, Steiner, Shamai and Diggavi (T-IT 2008) on the broadcast approach with expected distortion.

2.3 Timeliness: when to update

Owns age, freshness, and scheduling theory with bounds. Assumes away content (every update ages alike), the receiver's own sensing (partly addressed by Update the Unseen Only), and decision deadlines.

Table 3. Timeliness.

Paper	Venue	Note
Kaul, Yates, Gruteser	INFOCOM 2012	the origin of the age of information
Yates, Sun, Brown, Kaul, Modiano, Ulukus	JSAC 2021	survey
Kadota, Sinha, Uysal-Biyikoglu, Singh, Modiano	ToN 2018	AoI scheduling; Whittle index and max-weight
Sun, Polyanskiy, Uysal	T-IT 2020 [V]	signal-aware sampling beats age-optimal sampling
Maatouk, Kriouile, Assaad, Ephremides	ToN 2020	age of incorrect information
Fresh2comm (Wu, Peng, Yu)	arXiv 2502.07852, 2025 [S]	AoI for collaborative perception; author list from a secondary citation
TAMP (Zhu, Sun, Jia, Chen, Ai, Zhou)	arXiv 2601.04542, 2026 [V]	Lyapunov region scheduling with an age-to-accuracy penalty
AoI-guaranteed scheduling for CP (Wang/Xu, Sun, Zhou, Niu)	WiOpt 2023 [S]	centre-side resource allocation under age constraints; author list differs between the arXiv and WiOpt versions
Han, Wang, Liwang, Niyato	TCCN 2026 [V]	AoI derived from message size and link, weighting fusion
Update the Unseen Only (Ma, Zhao, Fang, An, Chen, Fang)	arXiv 2607.20967, 2026 [F]	age over areas users cannot sense; INAR persistence model; LocMW with sublinear regret
WiSwarm (Tripathi et al.)	INFOCOM 2023 [S]	AoI scheduling on a real multi-UAV Wi-Fi rig
Tripathi et al., computation and communication co-design	WiOpt 2021 [S]	joint compute and channel scheduling by AoI; occupancy-grid mapping

2.4 Mobile systems: who sends to whom on a real radio

Owns relevance, visibility, bandwidth partitioning and measured latency on real radios. Assumes away spoil times from content, decision-time metrics, and motion as a lever. AutoCast, EMP and AVR come from one line of work (Govindan; Mao), so a single group holds the systems ancestry.

Table 4. Mobile systems.

Paper	Venue	Note
AutoCast (Qiu, Huang, Asavisanu, Liu, Psounis, Govindan)	MobiSys 2022 [F]	per-object, per-receiver visibility from the current pose and relevance from trajectory collision; greedy by reward per byte; LTE-Direct TDMA; DSRC replay. The closest prior work
EMP (Zhang, Zhang, Sun, Zhu, Guo, Qian, Mao)	MobiCom 2021 [S]	edge-assisted multi-vehicle perception; bandwidth-adaptive point-cloud partition
AVR (Qiu, Ahmad, Bai, Gruteser, Govindan)	MobiSys 2018 [S]	augmented vehicular reality; vehicle-to-vehicle point clouds
Zhang et al., real-time multi-vehicle collaboration on asynchronous sensors	MobiCom 2023 [S]	asynchrony handled in real time
Wang, Cao, edge-assisted relevance-aware dissemination	IEEE 2024 [S]	send only objects whose trajectories intersect the receiver's
Lusvarghi et al., The Search for Relevance	TVT 2026 [V]	relevance theory for V2X content
V2X-ReaLO (Xiang et al.)	arXiv 2503.10034, 2025 [S]	online fusion over a live link
CooperScene	arXiv 2606.31219, 2026 [S]	per-scene C-V2X traces outdoors
SCooP-iT (this group)	in preparation	every link logged on the sensor clock and in the map frame, indoors, with replay validated; the only entry that supplies axis 4

2.5 Wireless access and resource allocation: who gets the channel

Owns user scheduling, slots, OFDMA, decentralised congestion control, learned access, and offloading. Assumes away what messages mean and when they are needed.

Table 5. Wireless access and resource allocation.

Paper	Venue	Note
Bianchi, performance analysis of the 802.11 DCF	JSAC 2000	the CSMA model
Heusse, Rousseau, Berger-Sabbatel, Duda, performance anomaly	INFOCOM 2003	slow senders occupy the airtime
IEEE 802.11ax, 802.11be	standards	uplink OFDMA, trigger frames, buffer status reports, restricted target wake time
ETSI TS 103 324, TR 103 562	standards	CPM generation and redundancy rules; decentralised congestion control
Lyu et al., value-of-information prioritisation for CPM	VNC 2025 [S]	value of information in the standard
SchedCP	arXiv 2502.10456, 2025 [S]	deep RL user scheduling by CSI and semantics
PRISM (Mazandarani, Shokrnezhad, Taleb)	TNSM 2026 [F]	distributed multi-agent RL medium access over shared semantic segments
Yahya, Naeem, Alsisi, Butt, Ejaz, robust task offloading in DT-enabled UAV-IoT networks	IEEE OJ-COMS vol. 7, 2026, DOI 10.1109/OJCOMS.2026.3663770 [F]	MINLP offloading with box-uncertain positions; content-free tasks; orthogonal channels
Hady et al., multi-agent RL for resource allocation: a survey	AI Review 2025 [S]	the MARL line
Neely, Stochastic Network Optimization	2010	drift-plus-penalty
Kelly, Maulloo, Tan	JORS 1998	network utility maximisation

2.6 Communication-aware robotics: where to move for the link

Owns connectivity, throughput and radio maps as inputs to motion. Assumes away what is sent, the validity of content, and perception tasks.

Table 6. Communication-aware robotics.

Paper	Venue	Note
Muralidharan, Mostofi, Communication-aware robotics	Annu. Rev. Control Robot. Auton. Syst. 2021 [V]	the survey
Fink, Ribeiro, Kumar	Proc. IEEE 2012	robust control for mobility and wireless communication
Stephan, Fink, Kumar, Ribeiro	T-RO 2017	concurrent control of mobility and communication
Mostofi, Malmirchegini, Ghaffarkhah	ICRA 2010	channel estimation in robotic networks
FERMI (Luo et al.)	RSS 2025 [V]	multi-robot radio mapping
Amigoni, Banfi, Basilico	IEEE Intell. Syst. 2017	communication-restricted exploration survey
Gielis, Shankar, Prorok	Current Robotics Reports 2022	critical review of communications in multi-robot systems
Yang et al., online learning and connectivity maintenance	arXiv 2410.05798, 2024 [S]	connectivity-aware coordination

2.7 Multi-robot mapping and estimation: sharing a map

Owns novelty against a shared map, map increments, event triggering, and localisation against a prebuilt map. Assumes away deadlines, the channel, and dynamic objects.

Table 7. Multi-robot mapping and estimation.

Paper	Venue	Note
Rocha, Dias, Carvalho	ICRA 2005	entropy-utility measurement sharing
Kimera-Multi (Tian et al.)	T-RO 2022	bandwidth-parsimonious distributed SLAM
Swarm-SLAM (Lajoie, Beltrame)	RA-L 2024	sparse inter-robot sharing

Paper	Venue	Note
Commerge	arXiv 2606.25386, 2026 [S]	scan-subset exchange for merging
Trimpe, D'Andrea, event-based state estimation	TAC 2014	send on innovation; silence is information
Khronos (Schmid, Abate, Chang, Carlone)	RSS 2024	spatio-temporal metric-semantic SLAM with change
2Fast-2Lamaa (Le Gentil, Falque, Lisus, Barfoot)	IJRR 2026, arXiv 2410.05433 [F]	lidar-inertial localisation against a prebuilt map at under 16 % of a laptop CPU; free-space carving of dynamic objects; the cost of a preloaded static map on a robot

2.8 Networked control and co-design

Owns computation, channel and control under age of information. Assumes away perception content, real contended radios, and heterogeneous teams.

Table 8. Networked control and co-design.

Paper	Venue	Note
Tripathi et al., co-design for monitoring and control	WiOpt 2021 [S]	Whittle co-scheduling of processing and channel
Hahn, Stursberg, uncertain AoI in stochastic control	arXiv 2021 [S]	AoI as a control input
AoI-aware task-oriented scheduling for networked control on SDR	arXiv 2022 [S]	argues that AoI needs semantic measures
Park, Sahai et al., wireless network design for control systems	COMST 2018	survey

2.9 Caching and edge systems

Owns time-to-live, invalidation, prefetch and compute reuse. Assumes away validity from the environment and access patterns known from plans.

Table 9. Caching and edge systems.

Paper	Venue	Note
Belady	IBM Syst. J. 1966	optimal offline eviction
Fofack, Nain, Neglia, Towsley, TTL caches	venue to be checked [S]	TTL as a cache model
CrossRoI (Guo et al.)	MMSys 2021 [S]	cross-camera region-of-interest masks to cut compute and traffic
Spatula (Jain et al.)	SEC 2020 [S]	cross-camera correlations; 8–86 times less compute
Mobility-aware proactive caching at roadside units	surveys 2024–2025 [S]	prefetch by predicted trajectory

3. Where the Gaps Are, Read off the Categories

Table 10. The four axes of the scenario against the nearest prior work.

Axis	Nearest prior	What is missing
deadline from motion	AutoCast's time-to-collision relevance; AoI's age	a per-place deadline from the receiver's stopping distance
validity from the world	AoII (assumed source model); ETSI dynamics rules (per object)	a per-place change rate measured from revisits, used as the lifetime
need from the receiver's future	AutoCast's current visibility; Update the Unseen Only's statistical persistence	the receiver's own future view cast from its plan and the map
shared, short channel with content on it	mobile-systems rigs (no content model); medium-access work (no meaning); collaborative perception (no channel)	a measured contended access point where the content decision changes what arrives

Two consequences no category reports: a clairvoyant bound computed from the known paths and the logged link, and the fraction of the field's traffic that its receivers never needed.

4. Positioning

The computer-vision community fixes the channel, the communication-theory community fixes the content, the age-of-information community fixes the meaning, the mobile-systems community fixes the time, and the robotics community fixes the message; each obtains a clean problem by freezing what its neighbour studies. The scenario above leaves all four axes open at once, and the one quantity that ties them, AoII at the decision, is computable from state the agents already exchange. AutoCast is the paper to position against first; WiSwarm and Tripathi's co-design are the scheduling ancestors; EMP is the systems ancestor; Mostofi and Fink–Kumar are the motion ancestors; Trimpe's event triggering and ETSI's generation rules are the ancestors of sending on change.